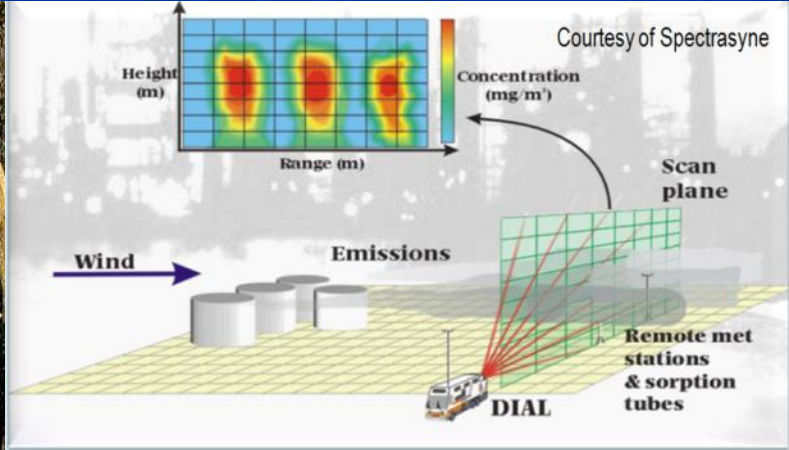
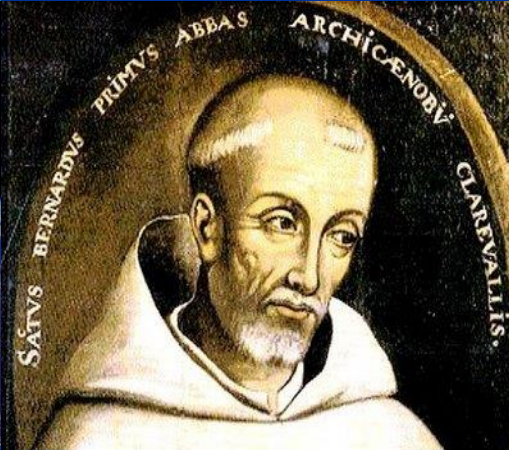
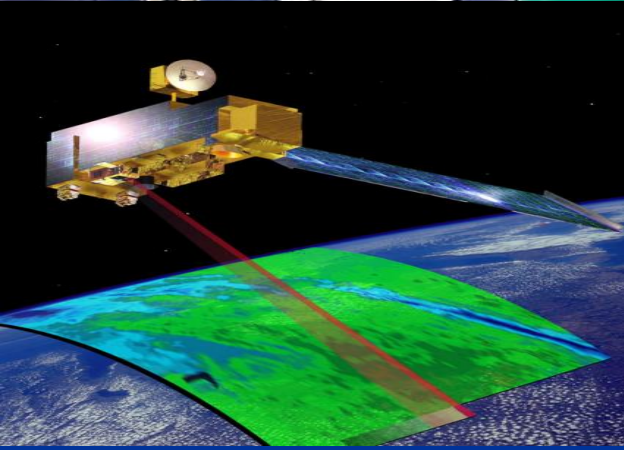




Occam's Razor and the Clean Air Act:

Can we and Should we Simplify the Clean Air Act?



A Take-Off on the Screwtape Letters:

How Can We Make More Money and Derive More Power from the Clean Air Act as Environmental Professionals?

1. Make the system more complicated

- The more complicated the system is, the more we are needed to help untangle the complexity for our clients and the more opportunities for arguments and litigation.

2. Make the system lengthier

- The lengthier the system is, the more opportunities for redundancies and conflict to result in opportunities for us, and the more difficult it is for non-environmental professionals to wrap their arms around what we are doing and assert themselves into our area of expertise.

3. Deny it is our responsibility to help change it

- We profit from the complexity and length of the CAA . . . but we didn't make it this way—nor is it our job to change it.

Who is More Likely to Hit a 90 MPH Fastball?



VS.



■ Why Update the Clean Air Act?

- System getting huge
- System getting more complex
- Dollars to manage going down
- Work going up
- System relying ultimately on States to achieve attainment—yet the ability of States to achieve attainment is decreasing
- New emissions measurement technologies are beginning to antique the current system that largely had to be built around emissions estimating techniques

**Roadmap of
Where We are
Headed Today**

■ Recent Developments and Advances on Clean Air Act Reform

- Breaking the Log Jam
- Energy & Commerce Committee - “Clean Air Act Forums”
- Canadian and British Reforming Clean Air Acts
- ECOS, TPPF, SIP Transformation Workgroup, and Other Efforts

■ Houston's Ability to Help Drive Clean Air Act Reform



The federal environmental statutes that Congress has addressed to EPA run to more than 2,700 pages in the two large, maroon-colored United States code volumes. The legally binding regulations issued by EPA to implement these statutes fill the 31 ocre-colored volumes of the Code of Federal Regulations. The guidance and other documents issued by EPA to explain or interpret its regulations fill around one million pages and are represented by the 1,250 grey-colored loose-leaf volumes. This does not include the millions of pages of State and local statutes, rules, and guidance that implement the millions of pages of Federal statutes, rules, and guidance.

COMPLEXITY OF THE CURRENT SYSTEM

Ex./ Regulations like those found in Part 63 are extremely complex and difficult for the public, regulators, and companies to understand.

LEGISLATIVE / EXECUTIVE / JUDICIAL

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(3) For each affected source complying with 63.802(a)(1) in accordance with 63.804(a)(3), compliance is demonstrated if: (i) The installation of a permanent total enclosure is demonstrated ($N=1$); (ii) The value of F is determined from Equation (5); and (iii) The product of $(FN)(100)$ yields a value (R) such that the value of E_{ac} in Equation 2 is no greater than 1.0.

(4) For each new affected source complying with 63.802(b)(1) in accordance with 63.804(d)(3), compliance is demonstrated if: (i) The installation of a permanent total enclosure is demonstrated ($N = 1$); (ii) The value of F is determined from Equation (5); and (iii) The product of $(FN)(100)$ yields a value (R) such that the value of E_{ac} in Equation 4 is no greater than 0.8.



“I hate that each sector has 17 to 20 rules that govern each piece of equipment and you’ve got to be a neuroscientist to figure it out”

--Gina McCarthy, EPA Administrator

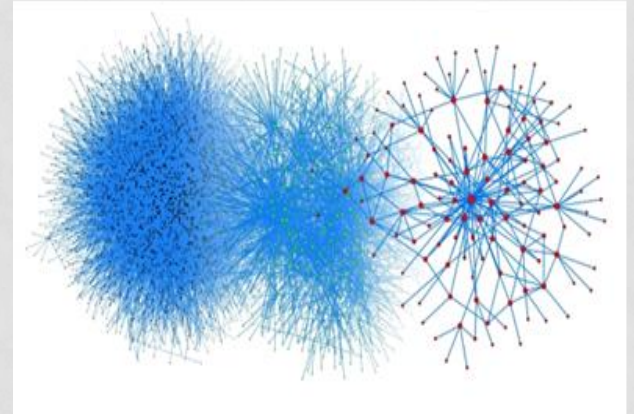
Other Comments About the Size and Complexity of the Current Clean Air Act

❖ **“The Clean Air Act is complicated and contentious”.** –Senate Environment and Public Works Committee

❖ **“The federal Clean Air Act alone has been referred to as the most complicated statute in history.”** --Erich Brich writing for the ABA

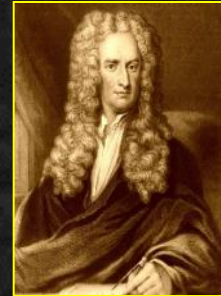
❖ **“The regulatory framework of the Clean Air Act is extraordinarily complicated. [. . .]The Act itself has often been called “unreadable” and “incomprehensible.”.** —John Quarles and Bill Lewis, Morgan & Lewis

❖ **“The statute and its regulatory offshoots are very complicated.”**
---U.S. Department of Justice

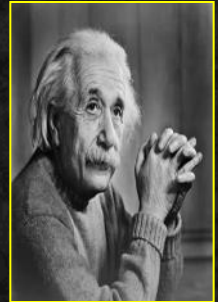


What do these people think about the complexity of the current Clean Air Act?

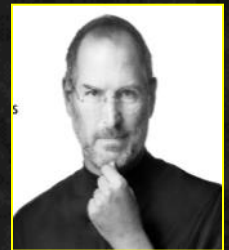
- “Truth is ever to be found in the simplicity, and not in the multiplicity and confusion of things.”---Isaac Newton



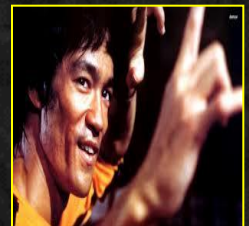
- “The definition of genius is taking the complex and making it simple.” ---Albert Einstein



- “Simple can be harder than complex: You have to work hard to get your thinking clean to make it simple. But it's worth it in the end because once you get there, you can move mountains. ---Steve Jobs



- “In building a statue, a sculptor doesn't keep adding clay to his subject. Actually, he keeps chiseling away at the inessentials until the truth of its creation is revealed without obstructions.”---Bruce Lee



- “Out of intense complexities, intense simplicities emerge.”---Winston Churchill

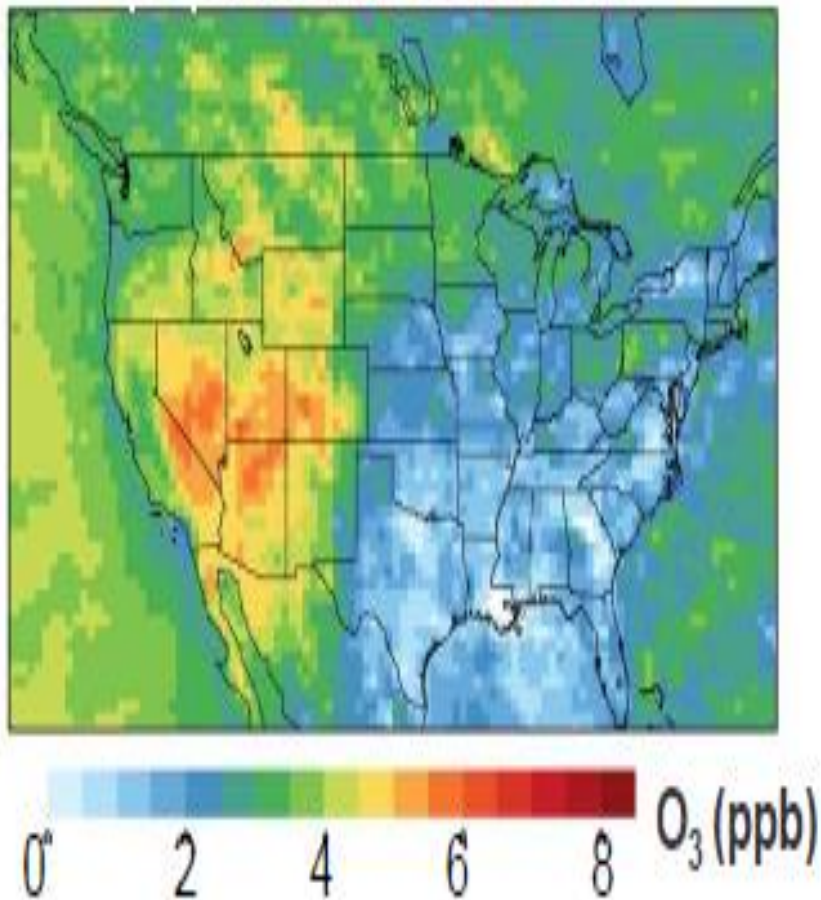


Ultimate Responsibility for Attaining the NAAQS Currently Rests with the States

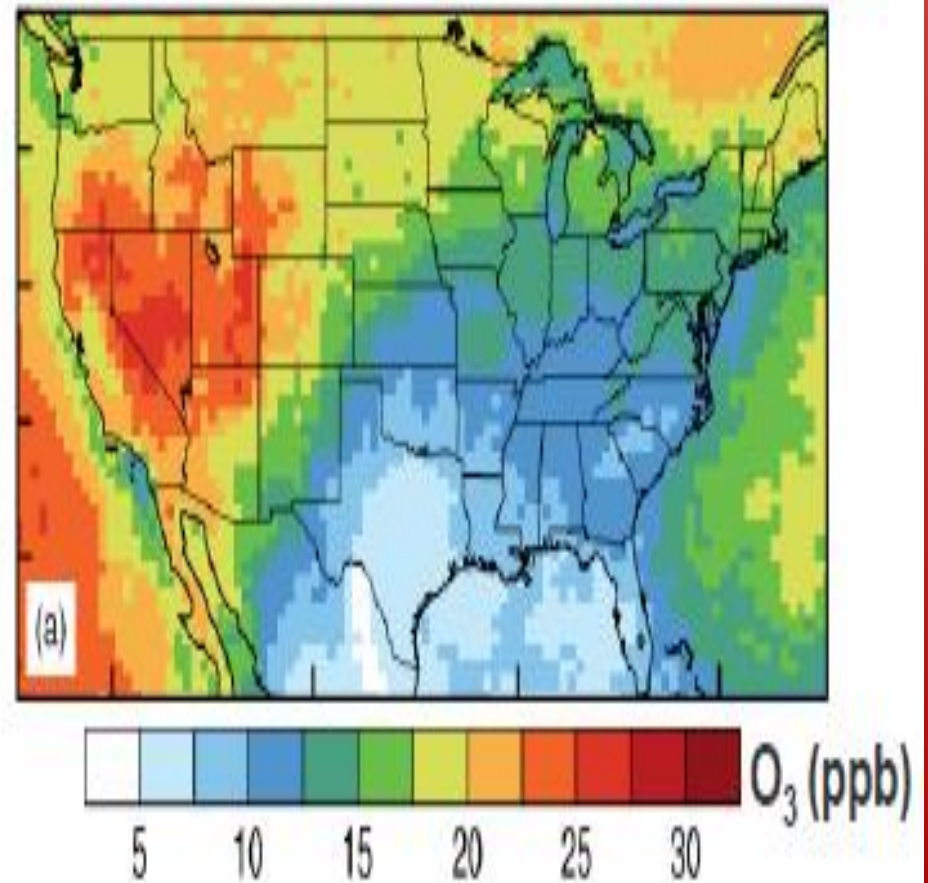
- **“Each State shall have the primary responsibility for assuring air quality within the entire geographic area comprising such state by submitting an implementation plan for such State which will specify the manner in which [the NAAQS] will be achieved and maintained . . .”**
-----Clean Air Act, 42 U.S.C. § 7407(a)
- **“Every State knows precisely what numerical goal its SIP must achieve. If a State misses that clear numerical target, it has only itself to blame.”**
-----D.C. Circuit Court of Appeals, CSAPR Ruling
- **“It is the States that the Act assigns initial and primary responsibility for deciding what emissions reductions will be required from which sources.”**
-----U.S. Supreme Court, American Trucking
- **If a State fails to address pollution in the air above its State it will be sanctioned by the Federal government.**
-----42 U.S.C. § 7509 and Section 185 of the CAA

Newly Discovered Impacts of Asian and Stratospheric Ozone Pollution Impacts on the U.S.

Asian: May-June 2010



Stratospheric (O_3S): April-June 2010

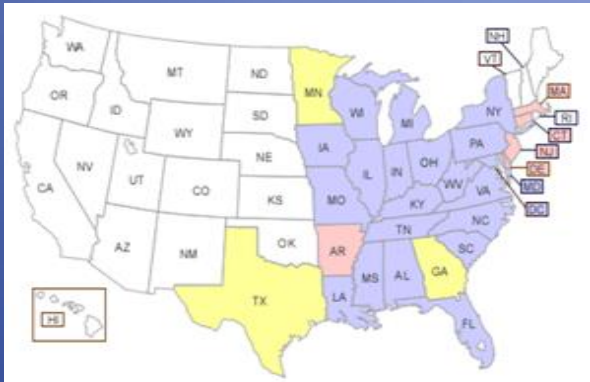


More and More Pollution is Outside Individual State/Local Control:

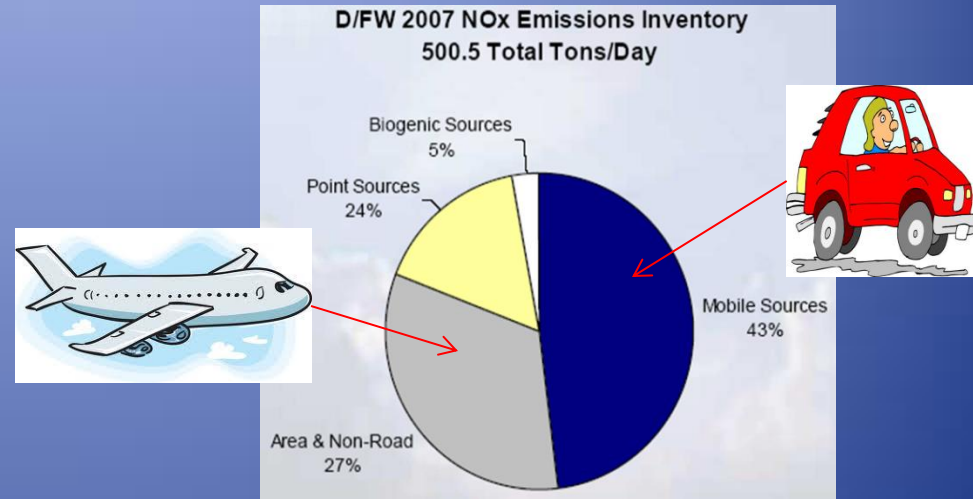
“It’s A Small Multi-Pollutant World After All”

Interstate Pollutant Transport

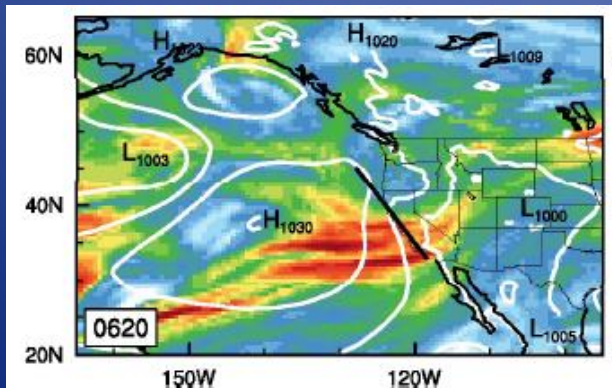
Ex. CAIR/CSAPR



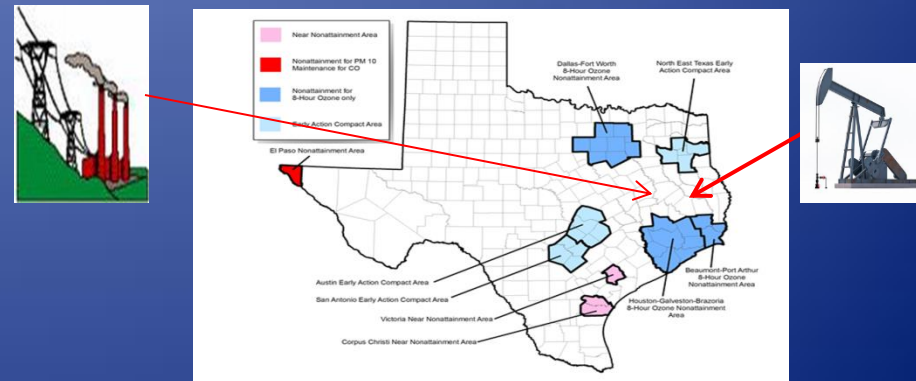
Federally Controlled Mobile Sources



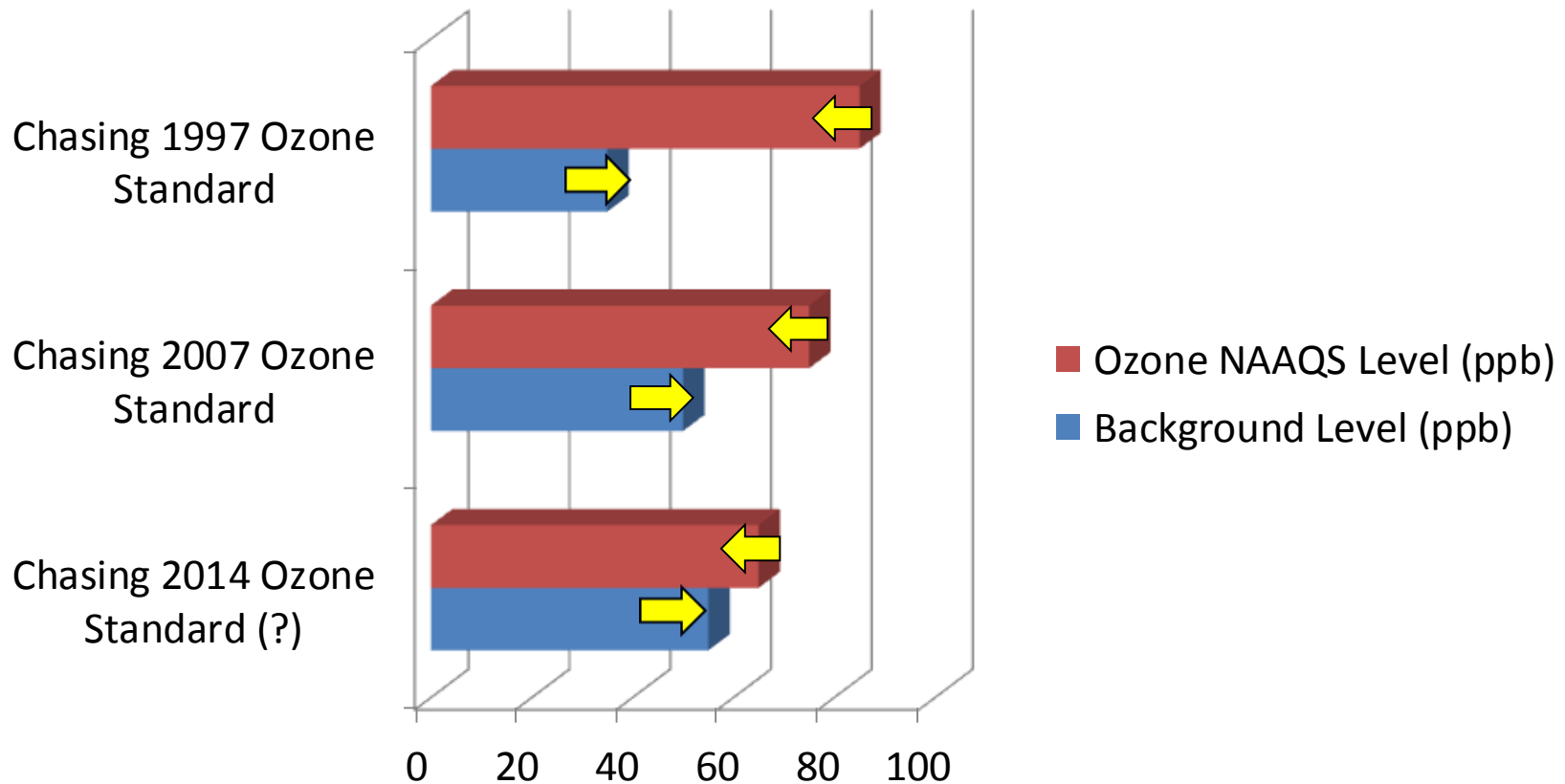
International Pollutant Transport



Intrastate Pollutant Transport



Difference Between Attainment and Nonattainment Areas is Disappearing—Along with the Corresponding Effectiveness of Making Such Geographical Distinctions for Air Quality Improvement Purposes



Nation is ultimately relying on the SIP process to achieve the NAAQS—yet the National Academy of Sciences said this about the process:

“The SIP process now mandates extensive amounts of local, state, and federal agency time and resources in a legalistic, and often frustrating, proposal and review process, which focuses primarily on compliance with intermediate process steps. This process probably discourages innovation and experimentation at the state and local levels; overtaxes the limited financial and human resources available to the nation’s AQM system at the state, local, and federal levels; and draws attention and resources away from the more germane issue of ensuring progress toward the goal of meeting the NAAQS.”

--- National Research Council (“Air Quality Management in the United States”, 2004)

Current System Based Largely on Estimating Emissions Due to the Historical Inability to Measure Most Emissions Sources

Emissions Factors in Reporting

All Reporting



Reporting using Indirect Methods

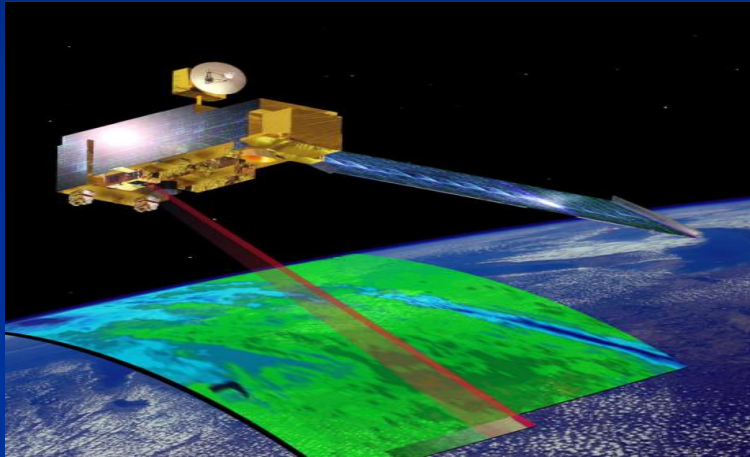


U.S. GAO, AIR POLLUTION: EPA SHOULD IMPROVE OVERSIGHT OF EMISSIONS REPORTING BY LARGE FACILITIES (2001)

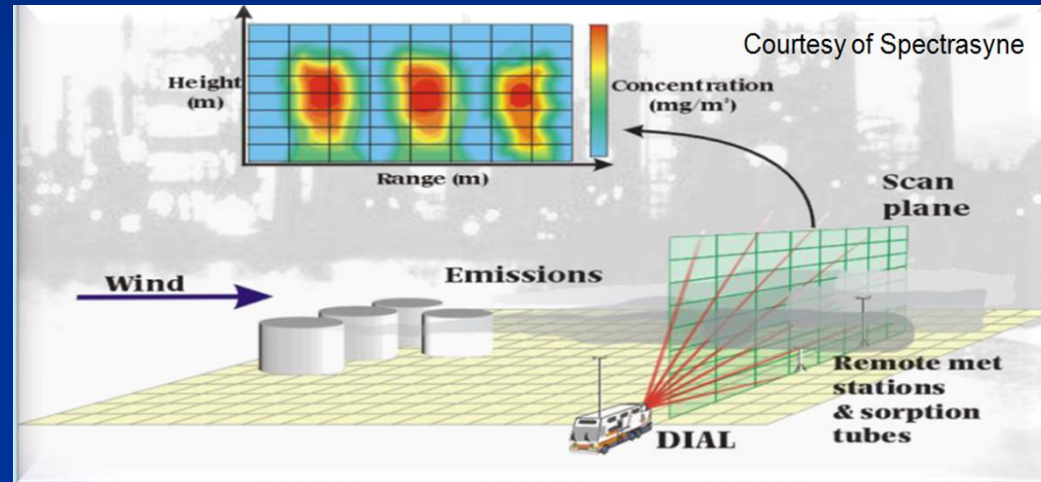
“Emission factors in AP-42 are neither EPA-recommended emission limits ... nor standards.... Use of these factors as source-specific permit limits and/or as emission regulation compliance determinations is not recommended by EPA.” ----U.S. EPA

Measurement Tools are Advancing that could Simplify, Improve, and Replace Current System

Satellite Monitoring



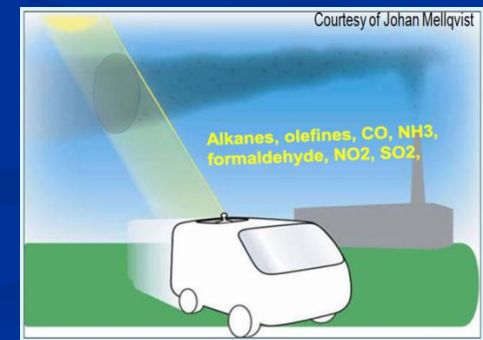
DIAL System



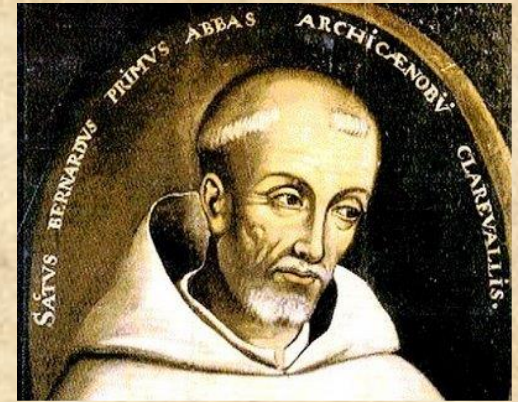
Fence-line Monitoring



SOF System



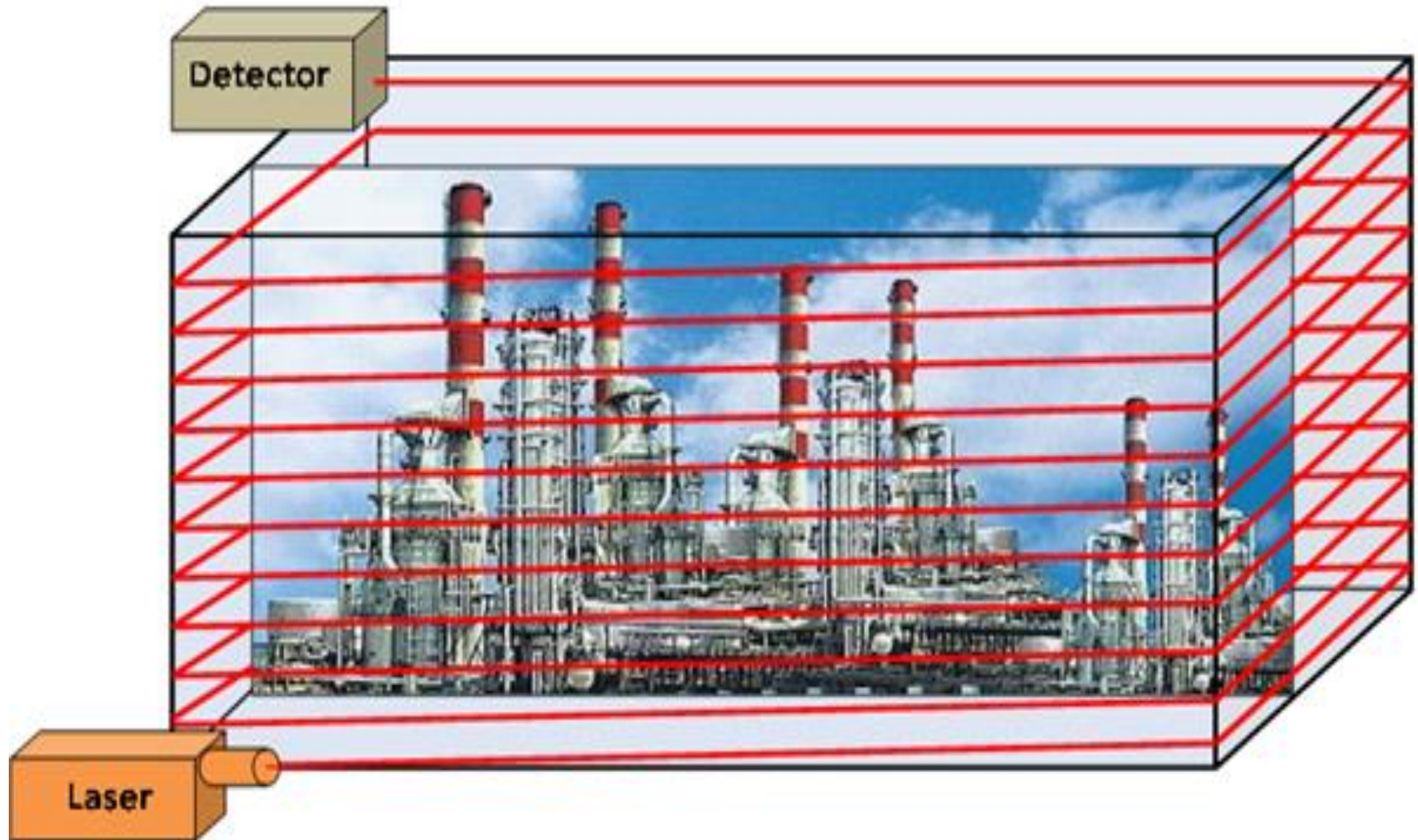
What would happen if we applied Occam's Razor to the Clean Air Act?



Occam Razor is that *“entities are not to be multiplied beyond necessity.”*

- ▶ **In physics:** Occam's Razor (or parsimony) was used to formulate the theory of special relativity by Einstein, the principle of least action by Mauepertuis and Euler, and quantum mechanics by Planck, Heisenberg, and Broglie.
- ▶ **In chemistry:** Theory of thermodynamics and the reaction mechanism.
- ▶ **In statistics and probability theory:** Kolmogorov complexity, Bayesian model comparison, and Laplace approximation

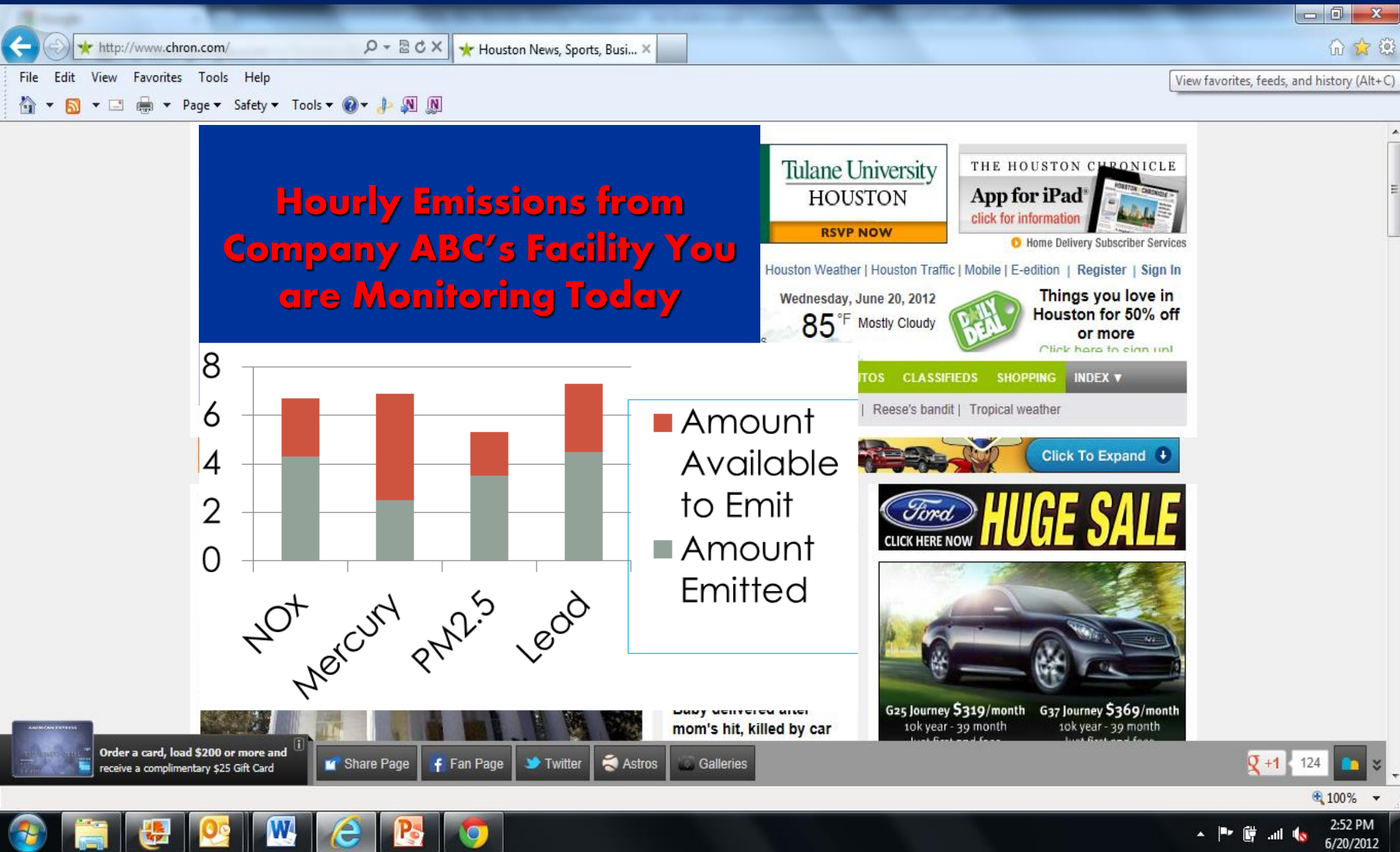
The Future?



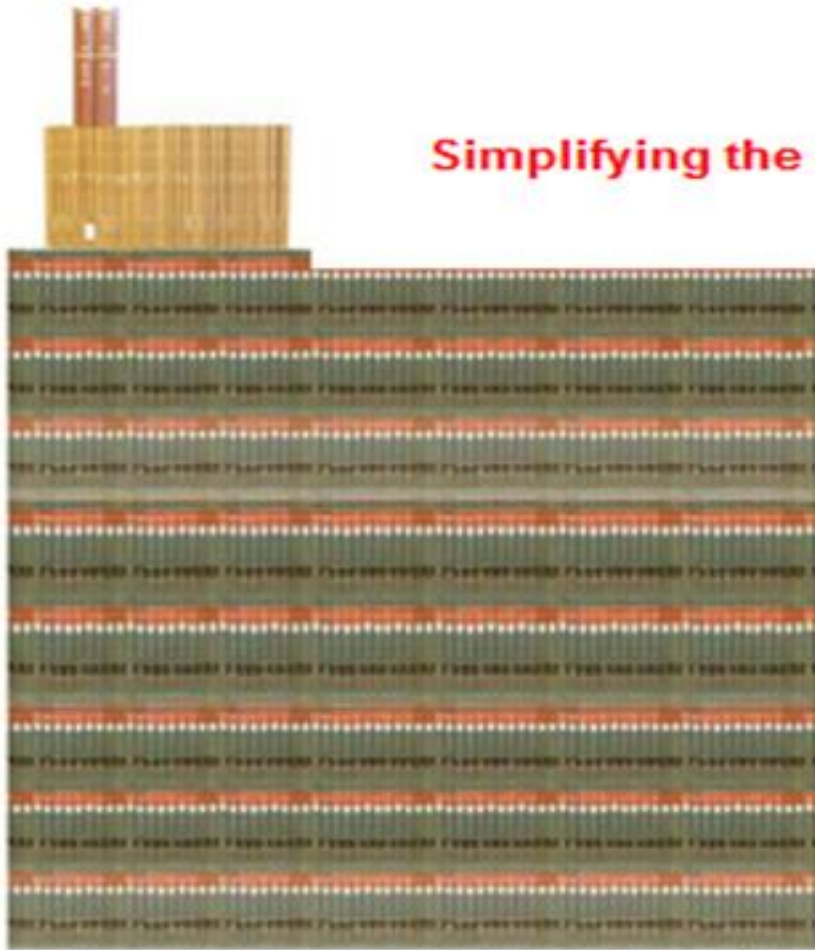
A photograph of an industrial facility, likely a refinery or chemical plant, at night. The facility is illuminated by numerous lights, creating a bright yellow glow against the dark blue sky. Several tall distillation columns and complex piping are visible. A large, yellow dashed arc is superimposed over the top half of the image, spanning from the left edge to the right edge, arching over the central part of the facility. In the background, a city skyline is visible across a body of water.

A company could do whatever it wanted
whenever it wanted within its facility so long as
the limits established by the monitoring/remote
sensing bubble were not exceeded

Imagine seeing this webpage as an environmental manager, regulatory agency, or a citizen instead of sifting through millions of pages of regulatory materials to determine if you or someone else is in compliance?



Simplifying the Air Quality Regulatory System



“Any intelligent fool can make things bigger, more complex, and more violent. It takes a touch of genius -- and a lot of courage -- to move in the opposite direction.”

-----E.F Schumacher

The Future Clean Air Act for Stationary Sources

PSD Program

Title V Program

NNSR Program

NSPS Program

MACT Program

Regional Haze Program

Acid Rain Program

CAIR/CSAPR Program

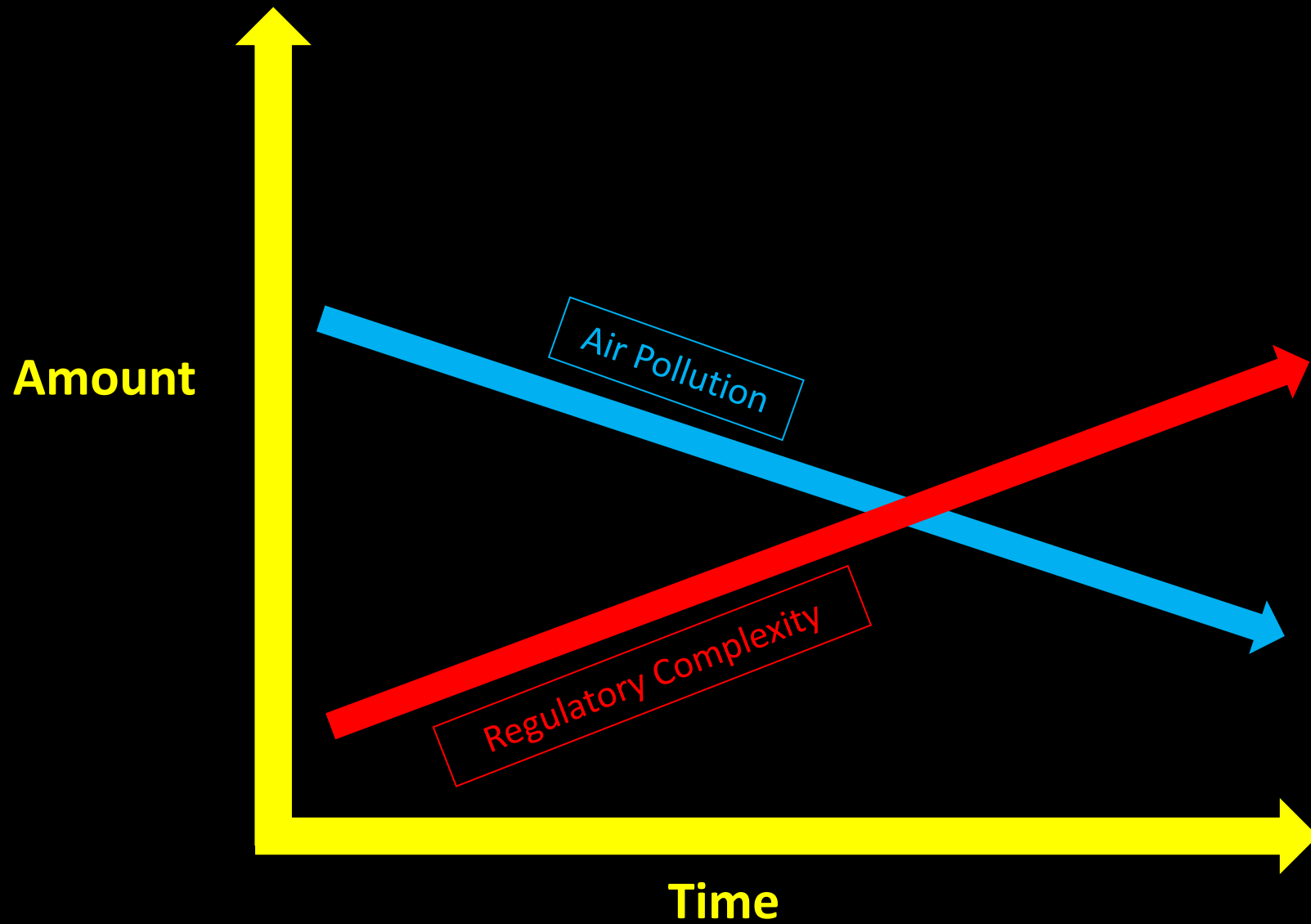
SIP requirements



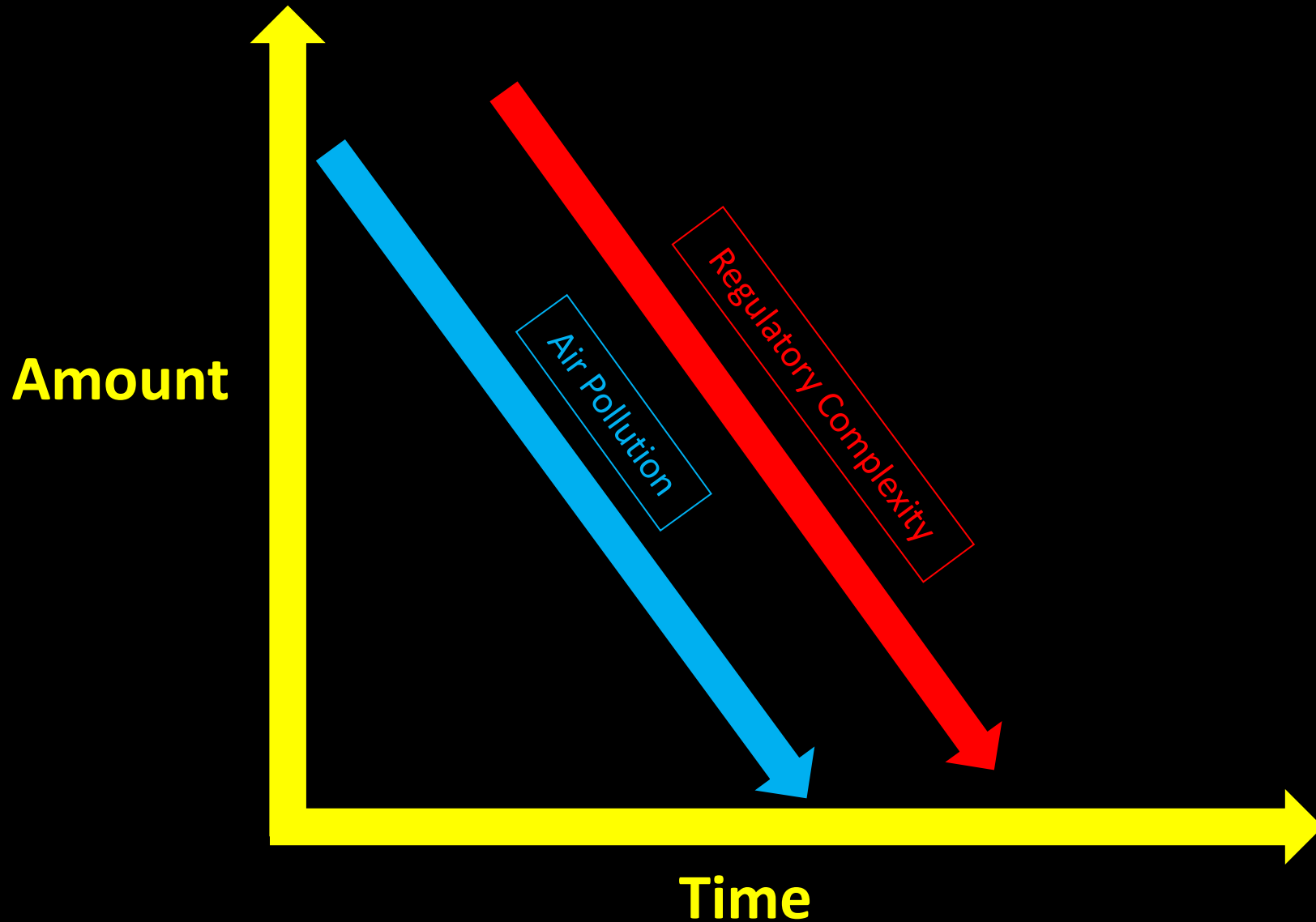
National/International Multi-Pollutant Market-Based System
Based on Real-Time Source
Monitoring

Approximately 50 to 75% of Clean Air Act regulations would no longer be needed.

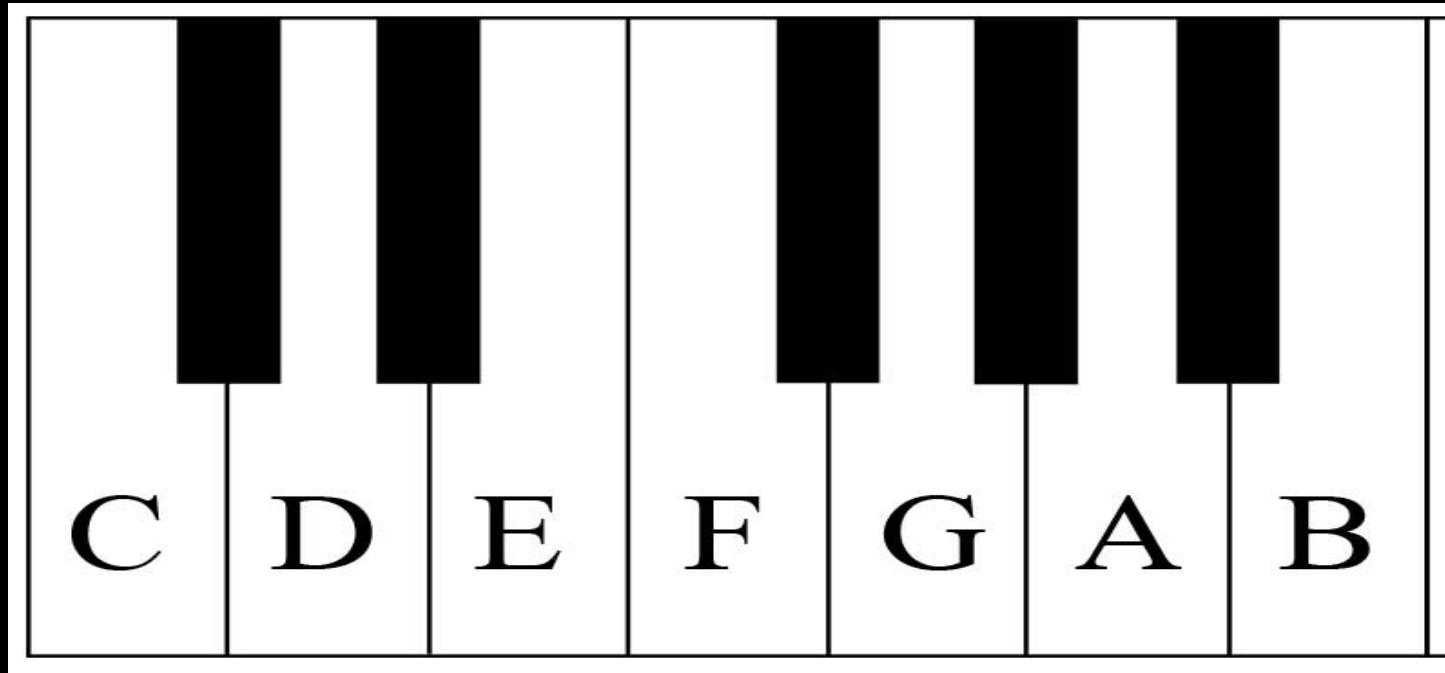
Current CAA Approach



New CAA Approach



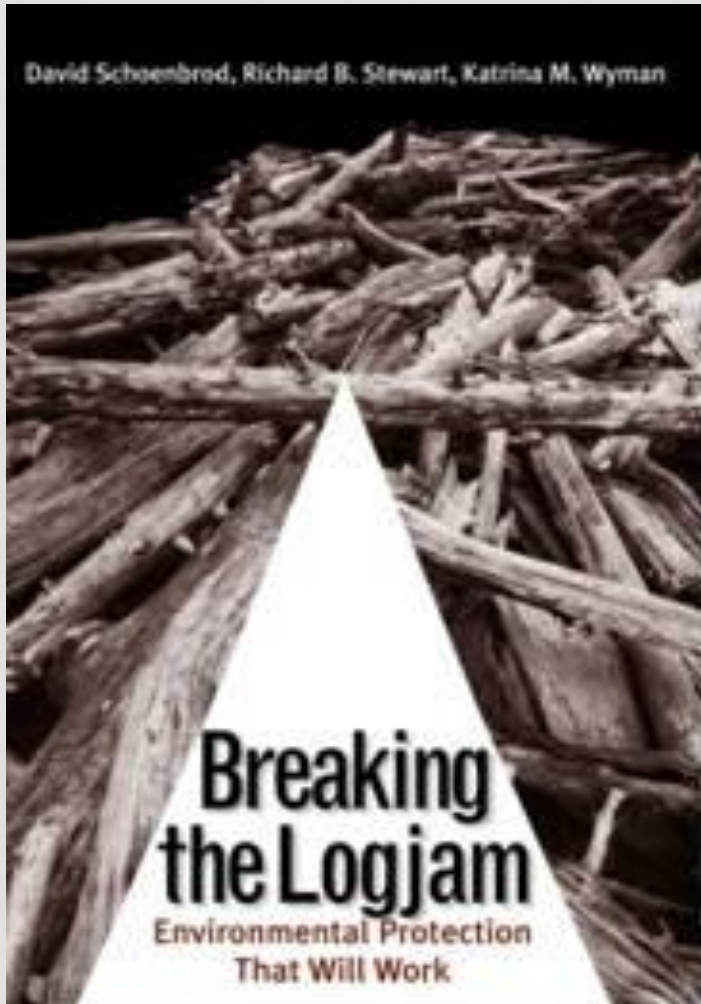
Think the CAA needs to remain long and complicated to deal with the complex and multifarious issues we now face?



Music only uses 12 notes. From Rachmaninoff to Bob Marley, from Muddy Waters to Eminem—only 12 notes have ever been used. Imagine what music we could play with a simplified Clean Air Act.

EFFORTS TO REFORM THE CLEAN AIR ACT

(EX. BREAKINGTHELOGJAM)



"Our environmental laws badly need re-thinking. This book makes that case and then charts a course for action. A straightforward, comprehensive and persuasive case for reform." —William D. Ruckelshaus, Former EPA Administrator

"The old environmental laws and approaches have run their course, and we urgently need an intensive period of environmental law reform. That's the message of this insightful, stimulating book. ...A must read." —James Gustave Speth, Member and Chairman of the White House Council on Environmental Quality

House Energy & Commerce Committee

- **State of Arkansas:** “The SIP process is **badly in need of reform**. The present process is overly cumbersome, slow and bureaucratic.”
- **State of Ohio:** “The State Implementation Plan process has become **burdensome and overly complicated**.”
- **State of Texas:** “While states are responsible for achieving the NAAQS through the SIP process, the **authority to achieve the ozone NAAQS arguably now lies with the federal government**. The nonalignment between responsibility and authority is a primary issue that needs to be considered in reform of the Clean Air Act.”
- **State of Colorado:** “SIPs and other demonstration packages from states are **only getting more voluminous and complex, often without concurrent air quality benefits**.”
- **San Joaquin Valley Air Pollution Control District:** “The current regiment [the NAAQS/SIP Process] leads to a **great deal of redundancy, overlap, and confusion**.”
- **South Coast Air Quality Management District:** “The current Clean Air Act places all the responsibility on the states, but then deprives them of the needed authority through preemption provisions. This is not a fair situation. **If USEPA has the sole authority, it must also have the responsibility**.”

Canada and Britain Updating Clean Air Acts



- The new **Canadian** system (2012) is called the “Canadian Air Quality Management System” (AQMS).
- System “zeros-out” transboundary air pollution blowing into a province from another country or from another province. [Under Canada’s new system the **provinces are now just responsible for pollution they can control**]
- According to Canada the new system is “designed to allow the **“best placed” government to act.**”



- Clean Air Act finding from an initial meeting of the **United Kingdom’s** Department for Environment, Food, and Rural Affairs:
 - “The overview from the meeting [. . .] was that the Act was **completely outdated** and **needed a full overhaul**. **All the terminology needed clarifying or modernizing**”
- UK’s Clean Air Act is 3 years newer than ours!

Still Think You are Too Small and Insignificant to Reform
the Clean Air Act? . . . Think Again.



Should We Change the Clean Air Act?

- ❖ "I cannot say whether things will get better if we change; what I can say is they must change if they are to get better." - Georg C. Lichtenberg
- ❖ "All conservatism is based upon the idea that if you leave things alone you leave them as they are. But you do not. If you leave a thing alone you leave it to a torrent of change." - G. K. Chesterton
- ❖ "The dogmas of the quiet past are inadequate to the stormy present. The occasion is piled high with difficulty, and we must rise with the occasion. As our case is new, so we must think anew and act anew." - Abraham Lincoln
- ❖ "If you have always done it that way, it is probably wrong." - Charles Kettering
- ❖ "I am not an advocate for frequent changes in laws and constitutions, but laws and institutions must go hand in hand with the progress of the human mind. As that becomes more developed, more enlightened, as new discoveries are made, new truths discovered and manners and opinions change, with the change of circumstances, institutions must advance also to keep pace with the times. We might as well require a man to wear still the coat which fitted him when a boy as civilized society to remain ever under the regimen of their barbarous ancestors." - Thomas Jefferson